```
import pandas as pd # pandas → used for data manipulation (DataFrame operations)

import numpy as np # numpy → used for numerical computations
```

```
df = pd.read_csv("titanic.csv") #pd.read_csv() loads the dataset into a DataFrame.
df.head() # head() shows first 5 rows to verify data loaded correctly.
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

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```
df.isnull().sum() # Returns count of missing values column-wise.
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df.describe() # Provides: Mean, Std deviation, Min/Max, Quantiles
```

```
df.describe() # Provides: Mean, Std deviation, Min/Max, Quartiles
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
df.info() # Shows:Number of entries, Data types, Non-null values
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.shape # Returns (rows, columns)
```

```
(891, 12)
```

```
df.dtypes #  Identifies variable types: int64 → integer, float64 → numeric, object → categorical/text
```

	0
PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object

dtype: object

```
# Ensures variables are in correct format for analysis.
df['Age'] = df['Age'].astype(float)
df['Survived'] = df['Survived'].astype(int)
```

```
# Min-Max Normalization scales values between 0 and 1
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()

# fit_transform(): fit() → learns min & max , transform() → scales data
df['Fare'] = scaler.fit_transform(df[['Fare']])
df['Age'] = scaler.fit_transform(df[['Age']])
```

```
# Converts categorical labels into numbers
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
```

```
# Creates separate columns: Embarked_C, Embarked_Q, Embarked_S
df = pd.get_dummies(df, columns=['Embarked'])
# avoid introducing artificial ordinal relationships, while numerical features are normalized using MinMaxScaler.
```

```
# mapping
```

```
df['Survived'] = df['Survived'].map({0: "Dead", 1: "Survived"})
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked_C	Embarked_Q	Embarked_S
0	1	Dead	3	Braund, Mr. Owen Harris	1	0.271174	1	0	A/5 21171	0.014151	NaN	False	False	True
1	2	Survived	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	0.472229	1	0	PC 17599	0.139136	C85	True	False	False
2	3	Survived	3	Heikkinen, Miss. Laina	0	0.321438	0	0	STON/O2. 3101282	0.015469	NaN	False	False	True
3	4	Survived	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	0.434531	1	0	113803	0.103644	C123	False	False	True

Next steps: Generate code with df New interactive sheet

1. What is Data Wrangling?

Process of cleaning, transforming, and organizing raw data into usable format.

2. What is Pandas?

A Python library used for data manipulation and analysis using DataFrames.

3. What is a DataFrame?

A 2D labeled data structure with rows and columns.

4. How do you load a dataset in Python?

1. pd.read_csv("file.csv")

2. from google.colab import files

uploaded = files.upload()

5. What is the purpose of head()?

Displays first 5 rows of dataset.

6. How do you check missing values?

df.isnull().sum()

7. What does describe() do?

Gives statistical summary (mean, std, min, max, quartiles)

8. What is info() used for?

Shows data types, non-null counts, memory usage

9. How do you find dataset dimensions?

df.shape

10. What are types of variables?

Numerical (int, float)

Categorical (object/string)

11. Why do we convert data types?

To ensure correct processing and analysis

12. What happens if data types are wrong?

Errors in computation and incorrect model results

13. What is normalization?

Scaling data to a fixed range (usually 0–1)

14. Why use normalization?

Prevents large values from dominating smaller ones

15. What is MinMaxScaler?

A tool from scikit-learn that scales data between 0 and 1

16. Formula of Min-Max Normalization?

17. What is categorical data?

Data representing categories (e.g., male/female)

18. What is Label Encoding?

Assigning numeric values to categories

19. What is One-Hot Encoding?

Creating binary columns for each category

20. Difference: Label vs One-Hot Encoding?

Label Encoding One-Hot Encoding

Assigns numbers Creates multiple columns

Introduces order No order introduced

21. Why not use LabelEncoder for all cases?

It creates false ordinal relationships

23. Which method is better for Embarked?

One-Hot Encoding

24. What if dataset has too many missing values?

Remove rows/columns or fill using mean/median/mode

25. What is feature scaling?

Standardizing range of independent variables

26. Difference between normalization and standardization?

Normalization Standardization

Range 0–1 Mean = 0, Std = 1

27. Why is preprocessing important?

Raw data is inconsistent and unsuitable for ML models

28. What is overfitting?

Model performs well on training but poorly on new data

29. Explain your entire workflow

Imported dataset, checked for missing values, analyzed structure using describe and info, corrected data types, normalized numerical features using MinMaxScaler, and converted categorical variables using encoding techniques to prepare data for machine learning.